PHYS 143 – Mechanics and Thermal Physics

# Work and Kinetic Energy

*How do forces change an object's motion as it moves through space?*

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## By the end of this week, you should be able to:

1. Calculate the work done by a constant force, and the total work done on an object.
2. Interpret the sign of work and the meaning of kinetic energy.
3. Apply the work–energy theorem to find speeds, distances and forces.
4. Calculate the work done by varying forces, including springs, and relate work to power.

**Central question**

How do forces change an object's motion as it moves through space?

**The thread of the week**

**force over a displacement → work → change in kinetic energy.** Find the work of every force, then use $W_{\text{tot}} = \Delta K$.

**Two contacts: A** (Tuesday) **IDENTIFY** when work is done and how much, then **QUANTIFY** how work changes motion with the work–energy theorem; **B** (Wednesday) **QUANTIFY** with a harder example, then **EXTEND** to varying forces, springs, curved paths and power.

**Reused:** the dot product (Week 1), areas under graphs and $v^2 = v_0^2 + 2a\Delta x$ (Week 2), free-body diagrams and friction (Week 3). **New:** work, kinetic energy, power. **Needed later:** Lectures 5, 7 and 8.

**Reading:** Young and Freedman, Chapter 6, §6.1–6.4. **Labels:** **CORE** you should be able to do this; **PRACTICE** develops the skill; **REFERENCE** keep for later.

# Where we start: a five-minute check

**Not marked. Answer on your own, then check.**

1. Find $\vec{A} \cdot \vec{B}$ for $\vec{A} = 3.0$ at $0^\circ$ and $\vec{B} = 4.0$ at $60^\circ$.
2. Find the area under a straight line from $(0, 0)$ to $(4, 6)$, and the area of a $3$ by $5$ rectangle.
3. A trotro speeds up uniformly from $10$ to $20\,\mathrm{m/s}$ over $50\,\mathrm{m}$. Find its acceleration.
4. A $10\,\mathrm{kg}$ box is pulled by a $50\,\mathrm{N}$ horizontal force against $20\,\mathrm{N}$ of friction. Find its acceleration.
5. A $5.0\,\mathrm{kg}$ block slides on a rough floor with $\mu_k = 0.30$. Find the friction force.
6. Convert $3000\,\mathrm{rev/min}$ to $\mathrm{rad/s}$.

**Answers**

1. $(3.0)(4.0)\cos 60^\circ = 6.0$
2. $\frac{1}{2}(4)(6) = 12$; $\quad 3 \times 5 = 15$
3. $(20^2 - 10^2)/(2 \cdot 50) = 3.0\,\mathrm{m/s^2}$
4. $(50 - 20)/10 = 3.0\,\mathrm{m/s^2}$
5. $0.30(5.0)(9.8) = 14.7\,\mathrm{N}$
6. $3000(2\pi/60) = 314\,\mathrm{rad/s}$

Item 1 is Week 1, items 2 and 3 are Week 2, items 4 and 5 are Week 3. All three return this week.

## Contact A: Identify, then Quantify

*When is work done, how much, and how does it change the motion?*

**Week 3**

$\sum \vec{F} = m\vec{a}$ at one instant. *What accelerates an object?*

**Week 4**

A force acting *through a displacement*. *What does it do to the speed?*

## Opening question

In which of these does the person do work **on the load**?
(a) A weightlifter holds a barbell motionless overhead.
(b) A kayayo walks along level ground at constant speed with a load on her head.
(c) A trader lifts a bag of rice from the floor onto a shelf.

*eg. kayayei, the head porters of Makola Market, Accra.*

## Predict first

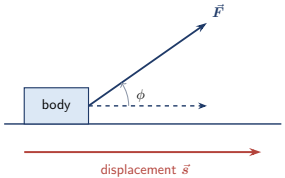
Decide first: in which cases does the load move, and how does the person's force point compared with the motion?

# The vocabulary of work and energy

CORE

| Quantity | What it is | What it tells you | SI unit | How to picture it |
|---|---|---|---|---|
| **Work** $W$ | energy transferred to an object by a force acting through a displacement | how much a force changed the object's energy of motion, and whether it gained or lost | joule, $1\,\mathrm{J} = 1\,\mathrm{N\,m}$ | area under the $F_x$–$x$ graph; sign from the angle $\phi$ |
| **Kinetic energy** $K$ | energy an object has because it is moving | the work needed to bring it from rest to its speed | J | a number, never negative |
| **Power** $P$ | rate at which work is done | how fast energy is transferred | watt, $1\,\mathrm{W} = 1\,\mathrm{J/s}$ | slope of a work–time graph |
| **Spring constant** $k$ | stiffness of a spring | force needed per metre of stretch | N/m | slope of the $F_x$–$x$ graph |

**Reading the table.** Work asks: *how much energy did this force give to the object, or take from it?* Kinetic energy asks: *how much work is stored in the motion?* We build these one at a time, starting with work.




**Key idea**

Only the part of $\vec{F}$ along $\vec{s}$ does work: $F\cos\phi$.

**Rule: work by a constant force**

$$W = \vec{F}\cdot\vec{s} = Fs\cos\phi \quad (1)$$

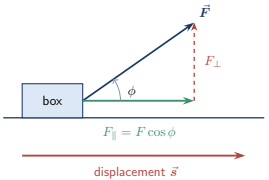
$$W = F_x s_x + F_y s_y \quad (2)$$

**Define.** $\vec{F}$ is the constant force on the object (N). $\vec{s}$ is the displacement of the object (m). $\phi$ is the angle between $\vec{F}$ and $\vec{s}$, measured tail to tail, from $0^\circ$ to $180^\circ$. $W$ is a **scalar** in joules, $1\,\mathrm{J} = 1\,\mathrm{N\,m}$.

Equation (2) is the same dot product written with components, as in Week 1.

**But what if...** the force is not along the motion? The next slide explains why only $F\cos\phi$ counts.

# Why only the component along $\vec{s}$ counts




**Key idea**

$F_\parallel$ changes the speed. $F_\perp$ only presses or lifts.

**Explain.**

**1.** Split $\vec{F}$ into a part along $\vec{s}$, $F_\parallel = F\cos\phi$, and a part across it, $F_\perp = F\sin\phi$.

**2.** Along the path, Newton's second law reads $F_\parallel = ma_\parallel$. So $F_\parallel$ is the only part that can speed the object up or slow it down.

**3.** $F_\perp$ changes the normal force or bends the path. It does not change the speed along the path, so it transfers no energy.

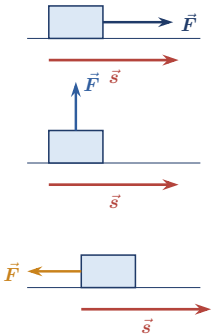
**4.** Hence $W = (F\cos\phi)\,s$, as in (1).

**Special cases.** $\phi = 0°$: $W = Fs$, the largest. $\phi = 90°$: $W = 0$. $\phi = 180°$: $W = -Fs$.

**Valid** for a constant force and a straight displacement of a point-like object.

# Predict the sign of work

CORE




$\phi = 0°$: $W > 0$
speeds up

$\phi = 90°$: $W = 0$
speed unchanged

$\phi = 180°$: $W < 0$
slows down

**Key idea**

The sign of $W$ is the sign of $\cos\phi$.

**Predict first**

A $10\,\mathrm{N}$ force acts through $3.0\,\mathrm{m}$. Predict $W$ for $\phi = 0°$, $60°$, $90°$ and $180°$ before the answers appear.

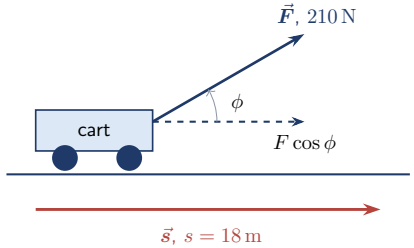
$\phi = 0°$ $\quad W = (10)(3.0)(1) = +30\,\mathrm{J}$
$\phi = 60°$ $\quad W = (10)(3.0)(0.50) = +15\,\mathrm{J}$
$\phi = 90°$ $\quad W = (10)(3.0)(0) = 0$
$\phi = 180°$ $\quad W = (10)(3.0)(-1) = -30\,\mathrm{J}$

**Rule: sign of work**

$0 \le \phi < 90°$: $W > 0$ (speeds up). $\quad \phi = 90°$: $W = 0$.
$90° < \phi \le 180°$: $W < 0$ (slows down).

Positive work gives energy to the object, negative work takes it away. A minus sign is not an error.

# Example 1: pulling a handcart




**Key idea**

Measure $\phi$ between the force and the displacement, not from the ground.

*eg. handcarts ("trucks") at Makola Market, Accra.*

**Example 1**

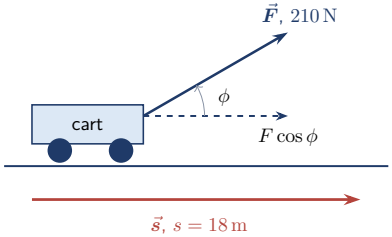
Ama pulls a loaded handcart $18\,\mathrm{m}$ along level ground. The rope tension is $210\,\mathrm{N}$, directed $30^\circ$ above the horizontal. How much work does the rope's force do on the cart?

**Predict first**

Positive, zero or negative? More or less than $Fs = 3780\,\mathrm{J}$?

# Example 1: pulling a handcart, solution

CORE




**Key idea**

Only $F\cos\phi = 182\,\mathrm{N}$ acts along $\vec{s}$. The rest lifts the cart slightly.

**IDENTIFY**

Given: $F = 210\,\mathrm{N}$, $s = 18\,\mathrm{m}$, $\phi = 30^\circ$ (force to displacement). Find: $W$ done by the rope.

**SET UP**

Constant force, straight path. Use (1): $W = \vec{F} \cdot \vec{s} = Fs\cos\phi$.

**EXECUTE**

Equation: $W = Fs\cos\phi$

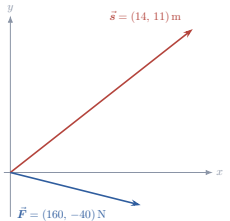
Substitute: $W = (210\,\mathrm{N})(18\,\mathrm{m})\cos 30^\circ$

Cosine: $\cos 30^\circ = 0.8660$, so $W = (3780\,\mathrm{N\,m})(0.8660)$

$$\boxed{W = 3.3 \times 10^3\,\mathrm{J}}$$

**EVALUATE**

Positive and less than $Fs = 3780\,\mathrm{J}$, as predicted. Units: $\mathrm{N\,m} = \mathrm{J}$. ✓




**Key idea**

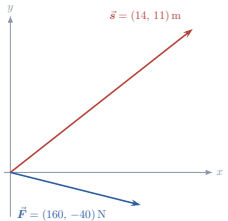
With components, multiply like with like: $x$ with $x$, $y$ with $y$.

**Independent problem**

The cart now goes up a ramp. A force $\vec{\boldsymbol{F}} = (160\,\mathrm{N})\,\hat{\boldsymbol{\imath}} - (40\,\mathrm{N})\,\hat{\boldsymbol{\jmath}}$ acts while its displacement is $\vec{\boldsymbol{s}} = (14\,\mathrm{m})\,\hat{\boldsymbol{\imath}} + (11\,\mathrm{m})\,\hat{\boldsymbol{\jmath}}$. Find $W$.

**Predict first**

The $y$-parts have opposite signs. Does the $y$-part add to $W$ or subtract?




**Key idea**

With components, multiply like with like: $x$ with $x$, $y$ with $y$.

**IDENTIFY**

Given: $F_x = 160\,\mathrm{N}$, $F_y = -40\,\mathrm{N}$, $s_x = 14\,\mathrm{m}$, $s_y = 11\,\mathrm{m}$.
Find $W$.

**SET UP**

Use (2), the component form: $W = F_x s_x + F_y s_y$.

**EXECUTE**

Substitute: $W = (160\,\mathrm{N})(14\,\mathrm{m}) + (-40\,\mathrm{N})(11\,\mathrm{m})$
Multiply: $W = 2240\,\mathrm{J} - 440\,\mathrm{J}$
$\boxed{W = 1.8 \times 10^3\,\mathrm{J}}$

**EVALUATE**

The $y$-part subtracts, as predicted. Check with (1):
$F = 165\,\mathrm{N}$, $s = 17.8\,\mathrm{m}$, $\phi = 52.2°$, so
$Fs\cos\phi = 1.8 \times 10^3\,\mathrm{J}$. ✓

# Concept check: which forces do no work?

**Does the named force do work? Give the sign, or zero.**

1. The weightlifter's force on the barbell while holding it overhead.
2. The kayayo's upward force on her load while she walks at constant speed on level ground.
3. The normal force of a table on a block sliding across it.
4. The tension in the string of a stone whirled in a horizontal circle.

**Predict first**

For each one, name $\vec{F}$ and $\vec{s}$ and decide the angle $\phi$ between them.

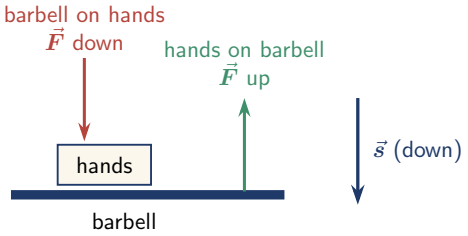
**Answer**

**All four do zero work.** (1) The barbell does not move, so $s = 0$. (2), (3), (4): the force is perpendicular to the displacement, $\phi = 90^\circ$ and $\cos 90^\circ = 0$.

**Common error:** "it takes effort, so work is done." Work needs a displacement with a component along the force. So in the opening question (a) is zero, (b) is zero and (c) is positive.

# Negative work and Newton's third law: lowering a barbell

CORE




## Key idea

The pair of forces is equal and opposite. Hands and barbell move together, so one work is $+$ and the other $-$.

## Predict first

A weightlifter lowers a $20\,\mathrm{kg}$ barbell vertically downward by $0.50\,\mathrm{m}$ at constant speed. Is the work done by the weightlifter on the barbell positive or negative? Is the work done by the barbell on the weightlifter's hands positive or negative?

**Explain.** Hands and barbell share the displacement $\vec{s}$ (down).

**Barbell on hands:** force down, along $\vec{s}$: $W = +Fs$.

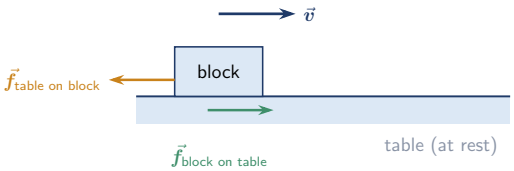
**Hands on barbell:** third law, $\vec{F}_{\text{hands on barbell}} = -\vec{F}_{\text{barbell on hands}}$, up and against $\vec{s}$: $W = -Fs$.

**Numbers:** constant speed gives $F = mg = 196\,\mathrm{N}$, so $W = \mp(196\,\mathrm{N})(0.50\,\mathrm{m}) = \mp 98\,\mathrm{J}$.

Negative work means the force opposes the motion; it keeps the barbell from crashing down.

# Do the works of a third-law pair always cancel?

CORE




## Decide first

A $2.0\,\text{kg}$ block slides $3.0\,\text{m}$ along a table, $\mu_k = 0.30$. The table does not move. Find the work done by each friction force of the pair, and their sum.

## Key idea

Equal and opposite forces cancel in work only if both objects have the same displacement.

## Predict first

Is the sum zero, as for the barbell?

**Block:** $f = \mu_k mg = 0.30(2.0)(9.8) = 5.88\,\text{N}$, opposite to $\vec{s}$, so $W = -(5.88\,\text{N})(3.0\,\text{m}) = -17.6\,\text{J}$.
**Table:** the force on the table is equal and opposite, but the table's displacement is zero, so $W = 0$.
**Sum:** $-17.6\,\text{J} \neq 0$. Where the two objects have different displacements the works need not cancel. That lost $17.6\,\text{J}$ warms the surfaces (Lecture 5 and thermal physics).

**Rule: total (net) work**

$$W_{\text{tot}} = W_1 + W_2 + W_3 + \cdots = (\sum \vec{\boldsymbol{F}}) \cdot \vec{\boldsymbol{s}} \tag{3}$$

**Define.** $W_i = \vec{\boldsymbol{F}}_i \cdot \vec{\boldsymbol{s}}$ is the work of force $i$ (J). $\sum \vec{\boldsymbol{F}}$ is the vector sum of all forces on the object (N). $W_{\text{tot}}$ is a scalar.

**Two methods.**
**Method I:** find each work with its sign, then add them (an *algebraic* sum).
**Method II:** find $\sum \vec{\boldsymbol{F}}$ first, then take $(\sum \vec{\boldsymbol{F}}) \cdot \vec{\boldsymbol{s}}$.

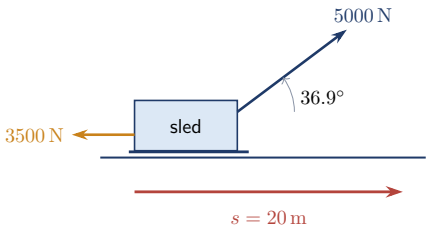
**Why they agree.** The dot product distributes over a sum: $(\vec{\boldsymbol{F}}_1 + \vec{\boldsymbol{F}}_2 + \cdots) \cdot \vec{\boldsymbol{s}} = \vec{\boldsymbol{F}}_1 \cdot \vec{\boldsymbol{s}} + \vec{\boldsymbol{F}}_2 \cdot \vec{\boldsymbol{s}} + \cdots$

**Predict first**

A box is dragged along a rough floor by a rope at an angle. Which forces do work, which do not, and which do negative work?

**Key idea**

List every force on the free-body diagram, give each work its sign, then add. Method II is the check. Valid for constant forces and a straight displacement.




**Key idea**

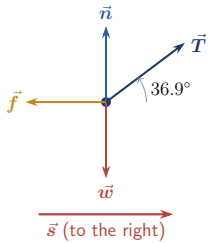
Draw the free-body diagram first, then give each force its own $\phi$ with $\vec{s}$.

**Example 2**

A farmer hitches her tractor to a sled loaded with firewood and pulls it $20\,\mathrm{m}$ along level ground. The total weight of sled and load is $14\,700\,\mathrm{N}$. The tractor exerts a constant $5000\,\mathrm{N}$ force at $36.9^\circ$ above the horizontal. A $3500\,\mathrm{N}$ friction force opposes the sled's motion. Find the work done by each force and the total work.

**Predict first**

Which forces do positive work, negative work, no work? Is the total positive?




**Key idea**

Weight and normal force are perpendicular to $\vec{s}$. Only the tractor and friction do work.

**IDENTIFY**

Given: $w = 14\,700\,\text{N}$, $T = 5000\,\text{N}$ at $36.9°$, $f = 3500\,\text{N}$, $s = 20\,\text{m}$. Find: $W$ of each force and $W_{\text{tot}}$.

**SET UP**

Forces: $\vec{T}$, $\vec{n}$, $\vec{w}$, $\vec{f}$. Constant forces, straight path: (1), $W = \vec{F} \cdot \vec{s} = Fs\cos\phi$. Total: (3).

Friction is given, so $n$ is not needed for the works.

**EXECUTE: Method I, add the works**

Weight and normal force ($\phi = 90°$): $W_w = W_n = 0$

Tractor: $W_T = Ts\cos 36.9° = (5000\,\text{N})(20\,\text{m})(0.800) = +8.0 \times 10^4\,\text{J}$

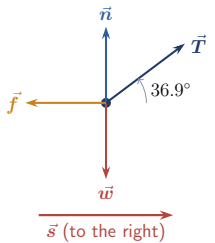
Friction ($\phi = 180°$):

$W_f = -fs = -(3500\,\text{N})(20\,\text{m}) = -7.0 \times 10^4\,\text{J}$

Total:

$W_{\text{tot}} = 8.0 \times 10^4\,\text{J} - 7.0 \times 10^4\,\text{J} = \boxed{1.0 \times 10^4\,\text{J} = 10\,\text{kJ}}$

# Example 2: a tractor pulls a sled, solution (2)




**Key idea**

The net force along $x$ is only $500\,\mathrm{N}$: the sled gains energy slowly.

**EXECUTE: Method II, check with the net force**

Equation (3): $W_{\text{tot}} = (\sum \vec{\boldsymbol{F}}) \cdot \vec{\boldsymbol{s}}$, with $\sum F_x = T\cos 36.9^\circ - f$

Substitute: $\sum F_x = 4000\,\mathrm{N} - 3500\,\mathrm{N} = 500\,\mathrm{N}$

Multiply: $W_{\text{tot}} = (500\,\mathrm{N})(20\,\mathrm{m}) = 1.0 \times 10^4\,\mathrm{J}$

Vertical check:

$n = w - T\sin 36.9^\circ = 14\,700\,\mathrm{N} - 3000\,\mathrm{N} = 11\,700\,\mathrm{N}$, so $\sum F_y = 0$

**EVALUATE**

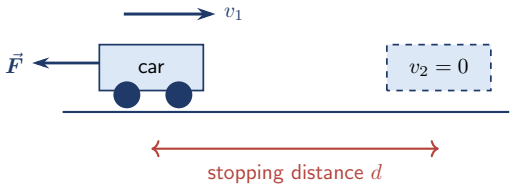
Signs as predicted: $W_T > 0$, $W_f < 0$, $W_w = W_n = 0$. Both methods give $10\,\mathrm{kJ}$. Positive, so the sled speeds up (next in this contact). Units: J. ✓

## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Calling effort "work". | Work needs a displacement with a component along the force. |
| Taking $\phi$ from the horizontal. | $\phi$ is the angle between $\vec{F}$ and $\vec{s}$, tail to tail. |
| Calculating before deciding the sign of $W$. | Decide the sign from $\phi$ first, then calculate. |
| Counting the normal force or weight on a level floor as doing work. | A force perpendicular to the displacement does none. |
| Using the work of one force as the total. | Add all the works, or use the net force. |

**But what if...** the total work is positive? What does that do to the object's speed? That is the next question.

# An opening question: twice the speed




**Poll**

A car on the Accra–Kumasi road, travelling at speed $v$, stops over a distance $d$ when the driver applies a constant braking force. The same car, travelling at $2v$, brakes with the **same force**. What is the stopping distance?

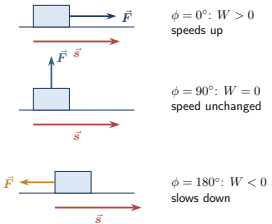
**Key idea**

Same car, same braking force. Only the starting speed changes.

**Predict first**

Twice $d$? Four times $d$? Eight times? Commit to an answer. We derive it, then check with numbers.

The first half of today told us that positive work speeds an object up and negative work slows it down. We now ask *by how much*.




## Key idea

The sign of the total work tells you whether the speed rises or falls.

## Idea

$W_{\text{tot}} > 0$: the object **speeds up**. $W_{\text{tot}} < 0$: it **slows down**. $W_{\text{tot}} = 0$: its speed **stays the same**.

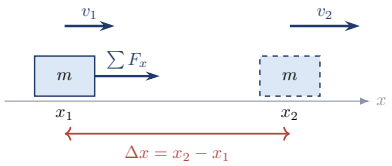
## Everyday cases

A footballer's kick on a ball rolling away: positive work. A goalkeeper's gloves stopping it: negative work. A ball held on a steady curve by a string: zero work.

## Predict first

A constant net force acts along the motion. Which quantity connects the force, the distance and the speeds? (Hint: Week 2, $v^2 = v_0^2 + 2a\Delta x$.)

# Deriving the work–energy theorem




**Key idea**

Newton's second law plus Week 2 kinematics give a link between $W$ and $v^2$.

**Setup.** A constant net force $\sum F_x$ acts along $+x$ on an object of mass $m$ as it moves from $x_1$ to $x_2$.

**Step 1.** Newton's second law: $\sum F_x = ma_x$

**Step 2.** Week 2: $v_2^2 = v_1^2 + 2a_x\Delta x$, so $a_x = \dfrac{v_2^2 - v_1^2}{2\Delta x}$

**Step 3.** Substitute into Step 1 and multiply by $\Delta x$:

$$\sum F_x \,\Delta x = m\,\frac{v_2^2 - v_1^2}{2} = \tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2$$

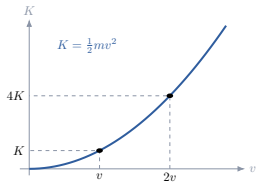
**Step 4.** The left side is $W_{\text{tot}} = (\sum F_x)\Delta x$ from (3) with $\phi = 0°$. Define $K = \frac{1}{2}mv^2$. Then $W_{\text{tot}} = K_2 - K_1$.

For a force that varies, and for a curved path, the same result holds (supplementary notes).

## Rule: work–energy theorem

$$K = \tfrac{1}{2}mv^2 \tag{4}$$

$$W_{\text{tot}} = K_2 - K_1 = \Delta K \tag{5}$$




**Key idea**

Doubling $v$ quadruples $K$.

**Define.** $m$ is the mass of the object (kg). $v$ is its speed (m/s). $K$ is its kinetic energy, a **scalar** that is never negative, in joules. $W_{\text{tot}}$ is the work of *all* forces on the object between state 1 and state 2.

**Meaning.** $W_{\text{tot}} > 0$ raises $K$, so the object speeds up. $W_{\text{tot}} < 0$ lowers $K$. $W_{\text{tot}} = 0$ leaves $K$ unchanged.

**Direction does not matter.** $K$ depends only on $m$ and $v$. Two identical balls at the same speed, one thrown up and one thrown east, have the same $K$. So (5) tells us about changes in *speed*, not velocity.

# Why the theorem uses net work, and why $K \propto v^2$

CORE

### Why the net work?

The derivation started from Newton's second law, $\sum F_x = ma_x$. That law holds for the *net* force, so (5) holds for the *total* work. Using the work of one force gives the wrong answer unless it is the only force that does work.

### Predict first

A crate is pushed with $50\,\mathrm{N}$ against $20\,\mathrm{N}$ of friction. Which work goes into (5): the push, the friction, or both?

**Both:** $W_{\text{tot}} = W_F + W_f$. Leaving out friction would overestimate $\Delta K$.

### Why $K \propto v^2$?

The squared speed came from Week 2, $v^2 = v_0^2 + 2a\Delta x$: each metre of constant push adds the same amount to $v^2$, and so the same amount to $K = \frac{1}{2}mv^2$. To go twice as fast you need four times the work.

$K = \frac{1}{2}mv^2$; if $v' = 2v$, then $K' = \frac{1}{2}m(v')^2 = \frac{1}{2}m(2v)^2$; $4\left(\frac{1}{2}mv^2\right) = 4K$.

### Valid for

A particle, meaning an object that moves without rotating. All forces count, including friction. It holds for varying forces too (supplementary notes).

## Decide, then compare with your neighbour

1. Can the total work on a moving object be zero while forces act on it?
2. Can kinetic energy be negative? Can work be negative?
3. The speed of an object doubles. What happens to its kinetic energy?
4. A force does negative work on an object. Does the object stop?
5. Two identical balls leave your hand at $10\,\mathrm{m/s}$, one up, one sideways. Compare $K$.

## Answers

**1. Yes.** At constant velocity the forces balance, or are perpendicular to the motion: $\Delta K = 0$.
**2.** $K$ **never;** $W$ **yes.** $K = \frac{1}{2}mv^2 \geq 0$. Negative work means the force opposes the displacement.
**3. It quadruples.** $K \propto v^2$.
**4. Not necessarily.** It slows down. Whether it stops depends on how much negative work there is compared with $K_1$.
**5. Equal.** $K = \frac{1}{2}mv^2$ has no direction.

### • METHOD – Applying the work–energy theorem

**1. STATES:** choose state 1 (start) and state 2 (end). Sketch both, with the speeds.
**2. FBD:** list every force on the object between 1 and 2, and draw $\vec{s}$.
**3. WORK:** find each $W$ with its sign, add for $W_{\text{tot}}$ (Eq. 3).
**4. WRITE** $W_{\text{tot}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$ and solve.
**WATCH OUT:** the net work, not one force; forces that do no work still belong on the diagram.
**CHECK:** does the sign of $\Delta K$ match speeding up or slowing down?

### Which method?

**Energy (5):** a *speed, distance or force* between two positions; no time needed. No time and no direction come out.

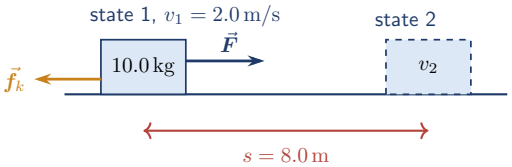
**Newton:** an *acceleration or a time*, or the direction matters.

### Decide: which method?

(a) Speed of a falling hammer at impact. (b) Time for a crate to slide to rest. (c) Force that stops a car in $40\,\text{m}$. (d) Acceleration on a slope.

**Answers:** (a) energy, (b) Newton, (c) energy, (d) Newton.

# Example 3: how fast after the push?




## Key idea

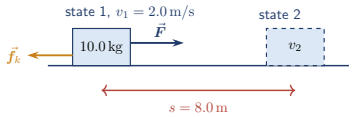
Mark both states on the sketch before you write any equation.

## Example 3

A trader slides a $10.0\,\mathrm{kg}$ crate of tomatoes along a market floor. It moves at $2.0\,\mathrm{m/s}$ when she starts pushing with a constant horizontal force of $50\,\mathrm{N}$ for $8.0\,\mathrm{m}$. Kinetic friction exerts $20\,\mathrm{N}$. What is its speed after the $8.0\,\mathrm{m}$?

## Predict first

Speeds up or slows down? About how much work does the net force do?




**Key idea**

$W_{\text{tot}} = \Delta K$ gives $v_2$ without $a$ or $t$.

## IDENTIFY

Given: $m = 10.0\,\mathrm{kg}$, $v_1 = 2.0\,\mathrm{m/s}$, $F = 50\,\mathrm{N}$, $f_k = 20\,\mathrm{N}$, $s = 8.0\,\mathrm{m}$. Find: $v_2$.

## SET UP

State 1: start of the push; state 2: after $8.0\,\mathrm{m}$. Use (5), $W_{\text{tot}} = K_2 - K_1$, with (3), $W_{\text{tot}} = \sum \vec{\boldsymbol{F}}_i \cdot \vec{\boldsymbol{s}}$.
$\vec{\boldsymbol{F}}$ along $\vec{\boldsymbol{s}}$; $\vec{\boldsymbol{f}}_k$ opposite; $\vec{\boldsymbol{n}}$ and $\vec{\boldsymbol{w}}$ perpendicular.

## EXECUTE

Push: $W_F = Fs\cos 0^\circ = (50\,\mathrm{N})(8.0\,\mathrm{m}) = +400\,\mathrm{J}$
Friction: $W_f = f_k s\cos 180^\circ = -(20\,\mathrm{N})(8.0\,\mathrm{m}) = -160\,\mathrm{J}$;
$W_n = W_w = 0$
Total: $W_{\text{tot}} = 400\,\mathrm{J} - 160\,\mathrm{J} = 240\,\mathrm{J}$
Start: $K_1 = \frac{1}{2}mv_1^2 = \frac{1}{2}(10.0\,\mathrm{kg})(2.0\,\mathrm{m/s})^2 = 20\,\mathrm{J}$
End: $K_2 = K_1 + W_{\text{tot}} = 260\,\mathrm{J}$, so $v_2 = \sqrt{2(260\,\mathrm{J})/10.0\,\mathrm{kg}}$

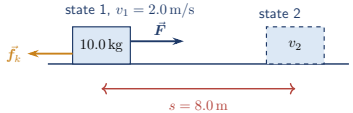
$\boxed{v_2 = 7.2\,\mathrm{m/s}}$

## EVALUATE

$W_{\text{tot}} > 0$ and $v_2 > v_1$, as predicted. Units: $\sqrt{\mathrm{J/kg}} = \mathrm{m/s}$. ✓

# Choosing the method: Example 3 again, by Newton's second law

CORE




**Key idea**

Same answer, different route. Newton also gives the time, at the cost of more steps.

**IDENTIFY**

Same data as Example 3. Find: $v_2$, and also the time $t$.

**SET UP**

Newton: $\sum F_x = ma_x$ (Week 3). Constant acceleration: $v_2^2 = v_1^2 + 2a_x\Delta x$ and $v_2 = v_1 + a_x t$ (Week 2).

**EXECUTE**

Acceleration: $a_x = \dfrac{F - f_k}{m} = \dfrac{50\,\mathrm{N} - 20\,\mathrm{N}}{10.0\,\mathrm{kg}} = 3.0\,\mathrm{m/s^2}$

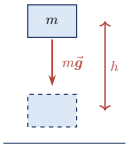
Speed: $v_2^2 = (2.0)^2 + 2(3.0)(8.0) = 52\,\mathrm{m^2/s^2}$ $\boxed{v_2 = 7.2\,\mathrm{m/s}}$

Time: $t = \dfrac{v_2 - v_1}{a_x} = \dfrac{7.21 - 2.0}{3.0}\,\mathrm{s}$ $\boxed{t = 1.7\,\mathrm{s}}$

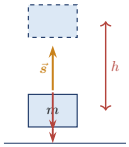
**EVALUATE**

$v_2 = 7.2\,\mathrm{m/s}$ agrees with the energy method. Only Newton gave $a_x$ and $t$; only the energy method skipped them. ✓

# Work done by the weight




moves down
$\phi = 0°$, $W_w = +mgh$




moves up
$\phi = 180°$, $W_w = -mgh$

## Key idea

Only the vertical change in height matters, not the path.

## Rule: work by gravity

$$W_w = \pm mgh \tag{6}$$

**State.** Moving **down** through $h$: $W_w = +mgh$. Moving **up**: $W_w = -mgh$.
**Define.** $m$ (kg); $g = 9.8\,\mathrm{m/s^2}$; $h > 0$ is the vertical height change (m).
**Explain.** This is (1) with $\vec{\boldsymbol{F}} = m\vec{\boldsymbol{g}}$ pointing down: $\phi = 0°$ going down, $180°$ going up. A horizontal step has $\phi = 90°$ and adds nothing.

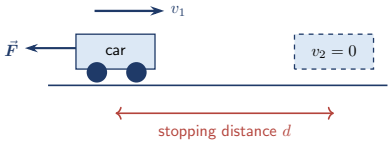
## Predict first

A stone goes up and falls back into the hand. Predict $W_w$ for the whole trip.

**Zero:** $-mgh$ up, $+mgh$ down. Lecture 5 builds potential energy on this.

# Example 4: back to the opening question, braking distance

CORE




**Key idea**

$d = mv_1^2/(2F)$: $d \propto v_1^2$ because $K \propto v_1^2$.

**Predict first**

Your poll answer: the distance at $2v$ is $4$ times the distance at $v$?

**Problem.** A $1200\,\text{kg}$ car at $25.0\,\text{m/s}$ stops with a constant braking force $F = 7200\,\text{N}$. Find $d$, then $d'$ from $12.5\,\text{m/s}$.

**IDENTIFY**

Given: $m = 1200\,\text{kg}$, $v_1 = 25.0\,\text{m/s}$, $v_2 = 0$, $F = 7200\,\text{N}$.
Find: $d$, and $d'$ for $v_1' = 12.5\,\text{m/s}$.

**SET UP**

(5), $W_{\text{tot}} = K_2 - K_1$. The brake force opposes $\vec{s}$, so $\phi = 180^\circ$ and $W_{\text{tot}} = -Fd$.

**EXECUTE**

Equation: $-Fd = 0 - \frac{1}{2}mv_1^2$, so $d = \dfrac{mv_1^2}{2F}$

Substitute: $d = \dfrac{(1200\,\text{kg})(25.0\,\text{m/s})^2}{2(7200\,\text{N})} = \dfrac{7.50 \times 10^5\,\text{J}}{1.44 \times 10^4\,\text{N}}$

$\boxed{d = 52\,\text{m}}$ Half speed: $d' = \dfrac{(1200)(12.5)^2}{2(7200)}\,\text{m} = \boxed{13\,\text{m}}$

**EVALUATE**

Matches Lecture 2 ($a = 6.0\,\text{m/s}^2$, $d = 52\,\text{m}$). Half the speed gives a quarter of the distance. ✓

## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Putting the work of one force into $W_{\text{tot}} = \Delta K$. | Use the total work of every force. |
| Thinking that doubling the speed doubles $K$. | $K \propto v^2$: it quadruples. |
| Using the energy method to find a time. | Use Newton's second law and kinematics. |
| Forgetting that friction removes kinetic energy. | Include $W_f = -f_k d$ with its sign. |
| Treating $K$ as a vector or as negative. | $K = \frac{1}{2}mv^2$ is a scalar and $\geq 0$. |
| Subtracting speeds instead of squares. | $\Delta K = \frac{1}{2}m(v_2^2 - v_1^2)$. |

### Contact A: what you can now do

- Find the work of a constant force, $W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}} = Fs\cos\phi$, decide its sign from $\phi$, and add the works of every force on a free-body diagram.
- State and use the work–energy theorem, $W_{\text{tot}} = \Delta K$ with $K = \frac{1}{2}mv^2$, to find a speed, a distance or a force without using time.
- Choose between the energy method and Newton's second law.

### Predict first

**Before Wednesday, predict:** a $200\,\text{kg}$ hammerhead falls $3.00\,\text{m}$ and then drives a beam $7.4\,\text{cm}$ into the ground. Is the average force on the beam larger or smaller than the hammer's weight? And what should we do when the force changes along the way, as the pull of a spring does?

**But what if...** the force changes along the way, as the pull of a spring does? Then $W = Fs\cos\phi$ with one value of $F$ cannot work. That is Wednesday: **varying forces, springs and power**.

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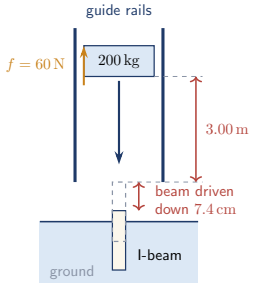
## Contact B: Quantify, then Extend

*How does the theorem handle a harder case,*
*and what changes when the force varies?*

**Predict first**

1. A $40\,\mathrm{N}$ force acts through a displacement of $5.0\,\mathrm{m}$, with $60^\circ$ between the force and the displacement. Find $W$. $W = Fs\cos\phi = (40)(5.0)\cos 60^\circ = 100\,\mathrm{J}$, positive.
2. A $4.0\,\mathrm{kg}$ cart moves at $3.0\,\mathrm{m/s}$. Find $K$, and the total work needed to double its speed. $K = \frac{1}{2}(4.0)(3.0)^2 = 18\,\mathrm{J}$. At $6.0\,\mathrm{m/s}$, $K = 72\,\mathrm{J}$, so $W_{\text{tot}} = \Delta K = 54\,\mathrm{J}$.
3. A net work of $+50\,\mathrm{J}$ acts on a $2.0\,\mathrm{kg}$ object that starts from rest. Find its speed. $50\,\mathrm{J} = \frac{1}{2}(2.0)v^2$, so $v = \sqrt{50} = 7.1\,\mathrm{m/s}$.

# Example 5: a pile driver with guide friction




## Example 5

At a building site the $200\,\mathrm{kg}$ steel hammerhead of a pile driver is lifted $3.00\,\mathrm{m}$ above the top of a vertical I-beam being driven into the ground. It is dropped and drives the I-beam $7.4\,\mathrm{cm}$ deeper. The vertical guide rails exert a constant $60\,\mathrm{N}$ friction force on the hammerhead. Find (a) the speed of the hammerhead just as it hits the I-beam, and (b) the average force the hammerhead exerts on the I-beam.

## Key idea

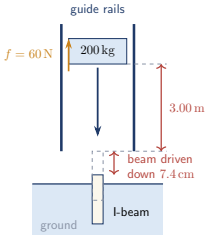
Two stages: free-ish fall (weight and rail friction) and the stop (weight, friction and the I-beam's force).

## Predict first

Does the $60\,\mathrm{N}$ matter much? Is the force on the I-beam larger or smaller than the weight, $1960\,\mathrm{N}$? About how many times?

# Example 5(a): the fall with rail friction

CORE




**Key idea**

Without friction $v = \sqrt{2gh} = 7.67\,\mathrm{m/s}$. The rails reduce it a little.

## IDENTIFY

Given: $m = 200\,\mathrm{kg}$, $s_{12} = 3.00\,\mathrm{m}$, $f = 60\,\mathrm{N}$, $v_1 = 0$. Find: $v_2$ just before impact.

## SET UP

State 1: at rest at the top. State 2: just before the I-beam. Weight $w = mg$ acts down, along $\vec{s}$; rail friction $f$ acts up, against $\vec{s}$. Use (3) and (5). *Reuse:* (1), (6)

## EXECUTE

Weight: $w = mg = (200\,\mathrm{kg})(9.8\,\mathrm{m/s^2}) = 1960\,\mathrm{N}$

Total work:

$W_{\text{tot}} = (w - f)s_{12} = (1960\,\mathrm{N} - 60\,\mathrm{N})(3.00\,\mathrm{m}) = 5700\,\mathrm{J}$

Theorem: $K_2 = W_{\text{tot}} = \frac{1}{2}mv_2^2$, so $v_2 = \sqrt{2(5700\,\mathrm{J})/200\,\mathrm{kg}}$

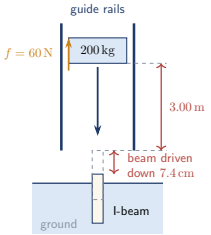
$\boxed{v_2 = 7.55\,\mathrm{m/s}}$

## EVALUATE

Slightly below the frictionless $7.67\,\mathrm{m/s}$, as expected. Units: $\sqrt{\mathrm{J/kg}} = \mathrm{m/s}$. ✓

# Example 5(b): driving the I-beam

CORE




**Key idea**

The I-beam stops $5700\,\mathrm{J}$ of motion in $7.4\,\mathrm{cm}$: a large force.

## IDENTIFY

Given: $m = 200\,\mathrm{kg}$, $f = 60\,\mathrm{N}$, $s_{23} = 0.074\,\mathrm{m}$, $K_2 = 5700\,\mathrm{J}$, $K_3 = 0$. Find: the average force $n$ of the I-beam.

## SET UP

(5) from contact (2) to rest (3). Weight along $\vec{s}$; friction $f$ and the I-beam's force $n$ against it. $n$ is the average.

## EXECUTE

Equation: $W_{\text{tot}} = (w - f - n)s_{23} = K_3 - K_2$

Solve:

$$n = w - f + \frac{K_2}{s_{23}} = 1960\,\mathrm{N} - 60\,\mathrm{N} + \frac{5700\,\mathrm{J}}{0.074\,\mathrm{m}} = 1900\,\mathrm{N} + 77\,000\,\mathrm{N}$$

$\boxed{n = 7.9 \times 10^4\,\mathrm{N}}$; by the third law the hammer pushes the I-beam with the same force.

## EVALUATE

About $40$ times the weight, as predicted. Whole-motion check: $n = (w - f)(s_{12} + s_{23})/s_{23} = 7.9 \times 10^4\,\mathrm{N}$. ✓

### What $K$ means

From Example 5(a): $K$ is the work done to bring an object **from rest to its speed**. From 5(b): $K$ is the work the object can do in being **brought to rest**.

### Predict first

Which has more kinetic energy: a $1000\,\mathrm{kg}$ taxi at $20\,\mathrm{m/s}$, or a $2000\,\mathrm{kg}$ trotro at $10\,\mathrm{m/s}$?

Taxi: $K = \frac{1}{2}(1000)(20)^2 = 2.0 \times 10^5\,\mathrm{J}$. Trotro: $K = \frac{1}{2}(2000)(10)^2 = 1.0 \times 10^5\,\mathrm{J}$. The taxi has **twice** the kinetic energy, although the trotro has twice the mass.

### Two iceboats, same force

Two iceboats of masses $m$ and $2m$ race on a frictionless frozen lake. The wind exerts the **same constant force** $F$ on each. Both start from rest and cross the finish line a distance $s$ away. Which has more kinetic energy at the finish?

Only $F$ does work, over the same $s$: $W_{\text{tot}} = Fs$ for both. They start from rest, so $K = Fs$ for both: **equal, whatever the mass.** The lighter boat is faster, $v_m/v_{2m} = \sqrt{2}$, and arrives sooner.

### Common error

Doubling the speed multiplies $K$ by $4$. Doubling the mass only doubles $K$ at equal speed.

**Question**

A $20\,\mathrm{kg}$ box of yams rests on the bed of a pickup truck. The truck accelerates at $2.0\,\mathrm{m/s^2}$ over $50\,\mathrm{m}$ and the box does not slide. Find the work done on the box by static friction, and compare it with the box's change in kinetic energy.

**Predict first**

Which way does friction point on the box? Is its work positive or negative?

**Answer**

$f_s = ma = (20)(2.0) = 40\,\mathrm{N}$, forward, along $\vec{s}$. So $W = +f_s s = +(40)(50) = +2000\,\mathrm{J}$.
The box starts from rest and ends with $v^2 = 2as = 200\,\mathrm{m^2/s^2}$, so $\Delta K = \frac{1}{2}(20)(200) = 2000\,\mathrm{J}$. ✓

**But what if...** you think friction always does negative work? It can do positive, negative or zero work, depending on $\phi$.

**Opening question**

A driver on the Accra–Kumasi road keeps changing the force on the car, pressing the accelerator, then the brake. We cannot use $W = Fs\cos\phi$ with one value of $F$. **How do we find the work?**

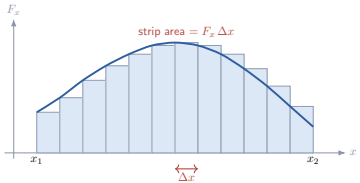
**Predict first**

Recall Week 2: how did we find the displacement when the velocity was not constant?

**The same idea, new quantity**

Cut the path into tiny steps $\Delta x$. Over each step $F_x$ is almost constant, so that step's work is $F_x\,\Delta x$. Add them up. For very small steps the sum becomes an integral.

*Reuse:* Week 2, area under the $v$–$t$ graph = displacement. Now the area under the $F_x$–$x$ graph = work.




**Key idea**

Each strip has area $F_x\,\Delta x$, a small work. The total area is $W$.

**Rule: work by a varying force**

$$W = \int_{x_1}^{x_2} F_x\,\mathrm{d}x \tag{7}$$

**Define.** $F_x(x)$ is the component of the force along the path (N), which may depend on position. $x_1$ and $x_2$ are the start and end *positions* (m), not times. $W$ is in joules.

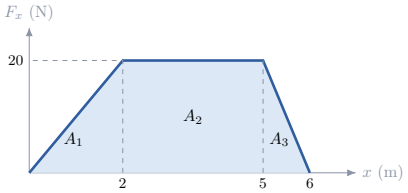
**Explain.** The integral is the sum of the strip works $F_x\,\Delta x$ as $\Delta x \to 0$. It equals the area between the curve and the $x$-axis. Area above the axis counts + (force along the motion). Area below counts − (force against the motion).

**Check.** If $F_x$ is constant, the area is a rectangle: $W = F_x(x_2 - x_1)$, which is (1) with $\phi = 0°$.

**Valid** for motion along a straight line with $F_x$ depending on position.

# Example 6: work from an $F_x$–$x$ graph

CORE




**Key idea**

Cut the shape into triangles and rectangles. Each area is a work in joules.

**Problem.** The force on a handcart varies with position as shown. Find the work it does from $x = 0$ to $x = 6.0\,\mathrm{m}$.
**Predict:** the largest force is $20\,\mathrm{N}$ over $6.0\,\mathrm{m}$. Is $W$ more or less than $120\,\mathrm{J}$?

**IDENTIFY**

Given: the graph. $F_x$ rises from $0$ to $20\,\mathrm{N}$ over $0$ to $2.0\,\mathrm{m}$, stays at $20\,\mathrm{N}$ to $5.0\,\mathrm{m}$, falls to $0$ at $6.0\,\mathrm{m}$. Find: $W$.

**SET UP**

(7): $W = \int F_x\,\mathrm{d}x =$ area. Split into triangle $A_1$, rectangle $A_2$, triangle $A_3$.

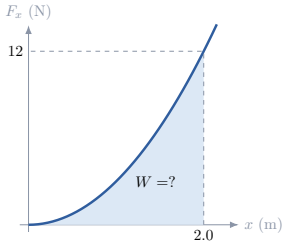
**EXECUTE**

$A_1 = \frac{1}{2}(2.0\,\mathrm{m})(20\,\mathrm{N}) = 20\,\mathrm{J}$
$A_2 = (3.0\,\mathrm{m})(20\,\mathrm{N}) = 60\,\mathrm{J}$
$A_3 = \frac{1}{2}(1.0\,\mathrm{m})(20\,\mathrm{N}) = 10\,\mathrm{J}$
$\boxed{W = A_1 + A_2 + A_3 = 90\,\mathrm{J}}$

**EVALUATE**

Less than $120\,\mathrm{J}$, as predicted. Units: $\mathrm{N\,m} = \mathrm{J}$. ✓

# Your turn: work from a formula




**Key idea**

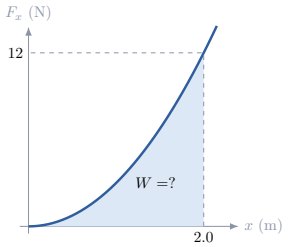
When $F_x$ is a formula, integrate. The graph is the check.

**Independent problem**

The force on a particle is $F_x = (3.0\,\mathrm{N/m^2})\,x^2$. Find the work done as it moves from $x = 0$ to $x = 2.0\,\mathrm{m}$.

**Predict first**

The force rises from $0$ to $12\,\mathrm{N}$. Is $W$ more or less than $12 \times 2.0 = 24\,\mathrm{J}$?




**Key idea**

When $F_x$ is a formula, integrate. The graph is the check.

**IDENTIFY**

Given: $F_x = 3.0x^2$ (SI units), $x_1 = 0$, $x_2 = 2.0\,\text{m}$. Find: $W$.

**SET UP**

$F_x$ varies with position, so use (7).

**EXECUTE**

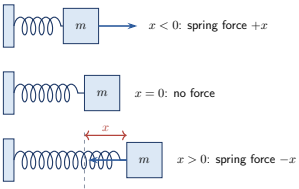
Equation: $W = \displaystyle\int_{x_1}^{x_2} F_x \, \mathrm{d}x$

Substitute: $W = \displaystyle\int_{0}^{2.0} 3.0x^2 \, \mathrm{d}x$

Integrate: $W = \Big[x^3\Big]_0^{2.0} = (2.0)^3 - 0$ $\qquad \boxed{W = 8.0\,\text{J}}$

**EVALUATE**

Less than $24\,\text{J}$, as predicted, because the force is small over most of the distance. ✓




**Key idea**

The spring always pulls or pushes back toward $x = 0$.

**Rule: Hooke's law**

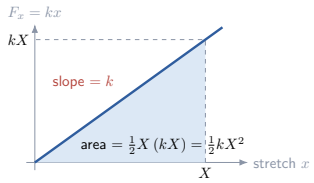
$$F_x = kx \tag{8}$$

**State.** The force *needed to hold* a spring stretched or compressed by $x$ is $F_x = kx$. The force *by the spring* on the object is $-kx$.

**Define.** $x$ is the displacement of the free end from its natural length (m), positive when stretched. $k$ is the spring constant (N/m): a stiff spring has a large $k$.

**Explain.** Doubling the stretch doubles the force, so $F_x$–$x$ is a straight line through the origin with slope $k$. The minus sign says the spring's force opposes the displacement: it is a *restoring* force.

**Valid** for small stretches. Beyond the elastic limit the line bends and the spring is permanently deformed.




## Key idea

The area is a triangle. Doubling $X$ makes the work four times larger.

## Rule: work on and by a spring

$$W_{\text{on}} = \tfrac{1}{2}kx_2^2 - \tfrac{1}{2}kx_1^2 \tag{9a}$$

$$W_{\text{by}} = \tfrac{1}{2}kx_1^2 - \tfrac{1}{2}kx_2^2 \tag{9b}$$

**Define.** $x_1$, $x_2$ are the start and end stretches, from natural length (m). $W_{\text{on}}$ is the work done on the spring by the force that stretches it; $W_{\text{by}}$ is the work done by the spring on the object attached to it.

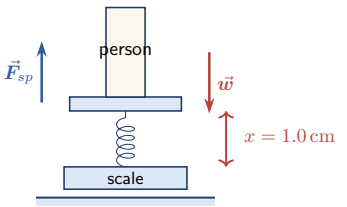
**Explain.** From (7) with $F_x = kx$: $W_{\text{on}} = \int_{x_1}^{x_2} kx\,\mathrm{d}x$, the area under the line. From $x = 0$ to $X$ it is a triangle of area $\frac{1}{2}X(kX) = \frac{1}{2}kX^2$. The spring pushes back, so $W_{\text{by}} = -W_{\text{on}}$.

## Predict first

Stretching from $0$ to $X$ takes $5.0\,\text{J}$. How much for $0$ to $2X$? ($20\,\text{J}$.)

# Example 7: stepping on a scale

PRACTICE




**Key idea**

At rest the spring force balances the weight: $kx = mg$.

**Problem.** Efua's $600\,\text{N}$ weight compresses a bathroom-scale spring by $1.0\,\text{cm}$. Find $k$ and the work done on the spring.

**Predict:** $k$ of order $10^2$, $10^4$ or $10^6\,\text{N/m}$? Work of order $1\,\text{J}$ or $100\,\text{J}$?

**IDENTIFY**

Given: $F = 600\,\text{N}$, $x = 1.0\,\text{cm} = 0.010\,\text{m}$, at rest. Find: $k$ and $W_{\text{on}}$.

**SET UP**

At rest $\sum F = 0$, so the spring force balances the weight: $F = 600\,\text{N}$. Hooke (8): $F = kx$. Work from $x_1 = 0$: (9a).

**EXECUTE**

Constant ($F = 600\,\text{N}$): $k = \dfrac{F}{x} = \dfrac{600\,\text{N}}{0.010\,\text{m}}$

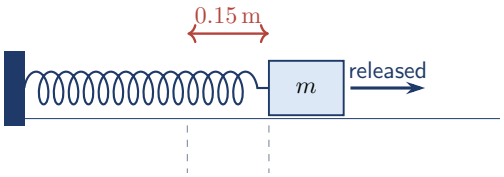
$$\boxed{k = 6.0 \times 10^4\,\text{N/m}}$$

Work: $W_{\text{on}} = \frac{1}{2}kx^2 - 0 = \frac{1}{2}(6.0 \times 10^4\,\text{N/m})(0.010\,\text{m})^2$

$$\boxed{W_{\text{on}} = 3.0\,\text{J}}$$

**EVALUATE**

Order $10^4\,\text{N/m}$ and about $3\,\text{J}$, as predicted. Check: $\frac{1}{2}Fx = 3.0\,\text{J}$. ✓




**Key idea**

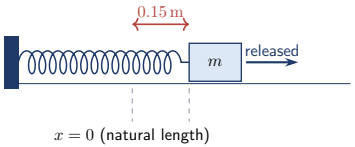
The spring force changes with $x$, but the energy method only needs the *work* between two positions.

**Example 8**

A $0.200\,\mathrm{kg}$ trolley on a laboratory air track is attached to a spring ($k = 40.0\,\mathrm{N/m}$). It is pulled $0.15\,\mathrm{m}$ from natural length and released from rest. Find its speed at $x = 0$ (a) with no friction, and (b) with a constant friction force of $0.50\,\mathrm{N}$.

**Predict first**

Is the speed in (b) larger or smaller than in (a)?




**Key idea**

Use (9b) for the spring and $-fd$ for friction, then (5).

**IDENTIFY**

Given: $m = 0.200\,\text{kg}$, $k = 40.0\,\text{N/m}$, $x_1 = 0.15\,\text{m}$, $v_1 = 0$, $f = 0.50\,\text{N}$ in (b). Find: $v_2$ at $x_2 = 0$.

**SET UP**

State 1: released at $x_1$. State 2: at $x = 0$. The spring force varies, so use (9b), $W_{\text{by}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$. Then (5). *Reuse:* (3), (5)

**EXECUTE**

Spring: $W_{\text{by}} = \frac{1}{2}(40.0\,\text{N/m})(0.15\,\text{m})^2 - 0 = 0.450\,\text{J}$

(a) $0.450\,\text{J} = \frac{1}{2}mv_2^2$, so $v_2 = \sqrt{\dfrac{2(0.450\,\text{J})}{0.200\,\text{kg}}}$ $\boxed{v_2 = 2.1\,\text{m/s}}$

(b) Friction: $W_f = -fx_1 = -(0.50\,\text{N})(0.15\,\text{m}) = -0.075\,\text{J}$

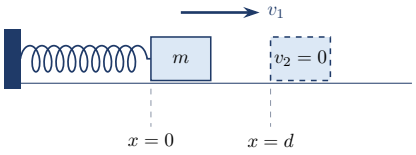
Total: $W_{\text{tot}} = 0.450\,\text{J} - 0.075\,\text{J} = 0.375\,\text{J}$, so

$v_2 = \sqrt{\dfrac{2(0.375\,\text{J})}{0.200\,\text{kg}}}$ $\boxed{v_2 = 1.9\,\text{m/s}}$

**EVALUATE**

(b) is smaller than (a), as predicted. Friction removed $0.075\,\text{J}$ of the $0.450\,\text{J}$. ✓

# Your turn: how far does the spring stretch?




## Key idea

At the largest stretch $v = 0$, so all the kinetic energy has gone into the spring and into friction.

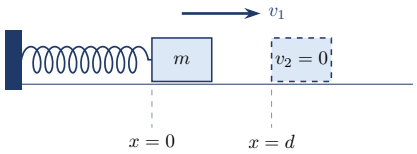
## Independent problem

The trolley of Example 8 starts at $x = 0$ (spring unstretched) moving to the right at $1.20\,\mathrm{m/s}$. Find the largest stretch $d$ (a) with no friction and (b) with a constant friction force of $0.50\,\mathrm{N}$.

## Predict first

Which stretch is larger, (a) or (b)? Which work is positive, which negative?

# Your turn: how far does the spring stretch?, solution




**Key idea**

At the largest stretch $v = 0$, so all the kinetic energy has gone into the spring and into friction.

**IDENTIFY**

Given: $m = 0.200\,\text{kg}$, $k = 40.0\,\text{N/m}$, $v_1 = 1.20\,\text{m/s}$ at $x_1 = 0$, $v_2 = 0$ at $x_2 = d$. Find: $d$.

**SET UP**

Spring: (9b), $W_{\text{by}} = -\frac{1}{2}kd^2$. Friction: $W_f = -fd$. Then (5), with $K_1 = \frac{1}{2}mv_1^2 = 0.144\,\text{J}$ and $K_2 = 0$.

**EXECUTE**

(a) $-\frac{1}{2}kd^2 = 0 - K_1$, so

$d = \sqrt{2K_1/k} = \sqrt{2(0.144\,\text{J})/40.0\,\text{N/m}}$ $\boxed{d = 0.085\,\text{m}}$

(b) $-\frac{1}{2}kd^2 - fd = -K_1$, so $20d^2 + 0.50d - 0.144 = 0$

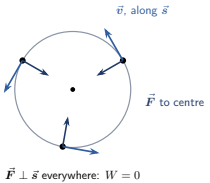
Quadratic formula:

$$d = \frac{-0.50 + \sqrt{0.25 + 11.52}}{40} = \frac{-0.50 + 3.43}{40}$$

$\boxed{d = 0.073\,\text{m}}$

**EVALUATE**

(b) is smaller than (a), as predicted. With $f = 0$ the quadratic reduces to (a). ✓




**Key idea**

$\vec{F} \perp \vec{s}$ at every point, so $W = 0$ and the speed does not change.

**Rule: perpendicular force**

A force that is perpendicular to the displacement at every point does **no work**.

**Explain.** Each small step has $\phi = 90°$, so $F\,\mathrm{d}s\cos 90° = 0$ and the sum is zero. Examples: the tension in a pendulum string, the normal force on a block sliding on a surface, the centripetal force in uniform circular motion.

Then $W_{\text{tot}} = 0$ and, by (5), $K$ and the speed are unchanged (Weeks 2 and 3).

**Limit.** Only the part of the force *along* the path counts. If a force also has a tangential part, that part does work and the speed changes.

**Decide first**

A $0.50\,\mathrm{kg}$ stone on a string moves in a horizontal circle of radius $0.80\,\mathrm{m}$ at a constant $2.0\,\mathrm{m/s}$. The tension is $T = mv^2/r = 2.5\,\mathrm{N}$, and one revolution is $2\pi r = 5.0\,\mathrm{m}$. What is the work done by the tension in one revolution?

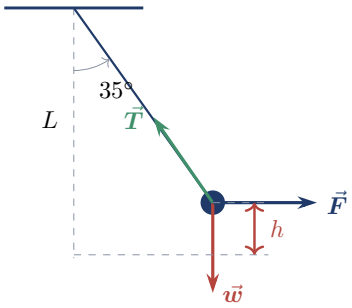
**Predict first**

Zero or non-zero? Compare the angle between $\vec{\boldsymbol{T}}$ and $\vec{\boldsymbol{v}}$ at each point.

**Answer**

**Zero.** $\vec{\boldsymbol{T}}$ is perpendicular to the velocity at every instant, so every small step has $\phi = 90°$. The product $T \times 2\pi r = 12.6\,\mathrm{J}$ is wrong, because $\vec{\boldsymbol{T}}$ and $\vec{\boldsymbol{s}}$ are not parallel. This agrees with $W = \Delta K = 0$: the speed is constant.

**But what if...** you also speed the stone up? Then the net force has a tangential part along $\vec{\boldsymbol{v}}$, and $W_{\text{tot}} = \Delta K \neq 0$.




**Key idea**

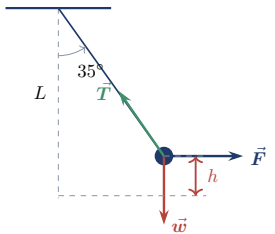
$\vec{T}$ is perpendicular to the path, so it does no work. Only $\vec{F}$ and $\vec{w}$ matter.

**Example 9**

A $25.0\,\mathrm{kg}$ child sits on a swing with chains $3.30\,\mathrm{m}$ long. You pull the child slowly with a horizontal force, so the child is almost in equilibrium at every point, until the chains make $35^\circ$ with the vertical. How much work do you do?

**Predict first**

Which of $\vec{F}$, $\vec{T}$ and $\vec{w}$ do work? What is $W_{\text{tot}}$?




**Key idea**

$F$ changes along the way, so $\int F\,ds$ is hard. The work–energy theorem avoids it.

**IDENTIFY**

Given: $m = 25.0\,\text{kg}$, $L = 3.30\,\text{m}$, $\theta = 35°$, slow motion. Find: $W_F$.

**SET UP**

States: hanging at rest, and at $35°$ at rest. So $K_1 = K_2$ and $W_{\text{tot}} = 0$ from (5). *Reuse:* (5), (6)

$W_F + W_T + W_w = 0$. Tension is perpendicular to the path: $W_T = 0$. Gravity: $W_w = -mgh$ (the child rises).

**EXECUTE**

Height: $h = L(1 - \cos\theta) = (3.30\,\text{m})(1 - 0.8192) = 0.597\,\text{m}$

Gravity:

$W_w = -mgh = -(25.0\,\text{kg})(9.8\,\text{m/s}^2)(0.597\,\text{m}) = -146\,\text{J}$

Solve: $W_F = -W_w - W_T = +146\,\text{J}$ $\boxed{W_F = 1.5 \times 10^2\,\text{J}}$

**EVALUATE**

Positive, as predicted: you push along the motion. It equals $mgh$, the work needed to raise the child through $h$. Units: J. ✓

## Rule: power

$$P_{\text{av}} = \frac{\Delta W}{\Delta t} = F_{\parallel} v_{\text{av}} \tag{10}$$

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}} \tag{11}$$

**Define.** $\Delta W$ is the work done (J) in the time $\Delta t$ (s). $F_{\parallel}$ is the component of $\vec{\boldsymbol{F}}$ along the path (N); $v_{\text{av}} = \Delta s/\Delta t$ is the average speed along it (m/s). $\vec{\boldsymbol{v}}$ is the velocity where $\vec{\boldsymbol{F}}$ acts. $P$ is a scalar in watts, $1\,\text{W} = 1\,\text{J/s}$.

**Explain.** Power is a *rate*. In (10), $\Delta W = F_{\parallel}\Delta s$, so $P_{\text{av}} = F_{\parallel}\,\Delta s/\Delta t = F_{\parallel} v_{\text{av}}$. For small $\Delta t$, $\mathrm{d}W = \vec{\boldsymbol{F}} \cdot \mathrm{d}\vec{\boldsymbol{s}}$ gives (11): $P = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}} = Fv\cos\phi$.

**Units.** $1\,\text{kW} = 10^3\,\text{W}$; $1\,\text{hp} = 746\,\text{W}$; a *kilowatt-hour* is energy: $1\,\text{kWh} = 3.6 \times 10^6\,\text{J}$.

*eg. ECG, the Electricity Company of Ghana, bills in kilowatt-hours.*

## Predict first

Two climbers lift the same load through the same height. One takes twice as long. Compare the work and the power.

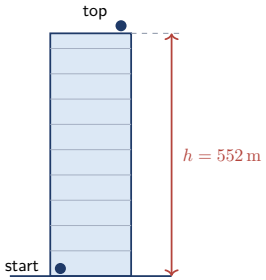
Same work; the faster climber has **twice the power**.

## Common error

Power is not energy. A kilowatt-hour is energy.

## Typical sizes

Person climbing: about $10^2$ W. Car cruising: about $10^4$ W.




**Example 10**

A $50.0\,\mathrm{kg}$ runner climbs the stairs of the $632\,\mathrm{m}$ Shanghai Tower, $552\,\mathrm{m}$ of climbing, in $25.0\,\mathrm{min}$. What must be her average power output, (a) from work and time, (b) from force and speed? Give it in W, kW and horsepower.

**Predict first**

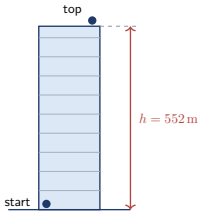
Will the answer be nearer $10$, $10^2$ or $10^4\,\mathrm{W}$? Do both methods agree?

**Key idea**

Two routes to the same power: work over time, or force along the path times average speed.

*eg. a stair race in a tall tower; Ghana has no building this tall, so the Shanghai Tower is kept.*

# Example 10: a power climb up a tower, solution




**Key idea**

Constant speed: the upward force equals her weight, $F_{\parallel} = mg$.

## IDENTIFY

Given: $m = 50.0\,\mathrm{kg}$, $h = 552\,\mathrm{m}$, $\Delta t = 25.0\,\mathrm{min} = 1500\,\mathrm{s}$.
Find: $P_{\mathrm{av}}$ in W, kW, hp.

## SET UP

Treat her as a particle lifted at constant speed. Method (a): $W = mgh$ (6), then $P_{\mathrm{av}} = \Delta W/\Delta t$ (10). Method (b): $P_{\mathrm{av}} = F_{\parallel} v_{\mathrm{av}}$ (10).

## EXECUTE

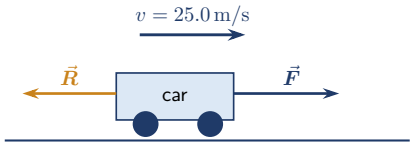
(a) Work:
$W = mgh = (50.0\,\mathrm{kg})(9.80\,\mathrm{m/s^2})(552\,\mathrm{m}) = 2.70 \times 10^5\,\mathrm{J}$
Power: $P_{\mathrm{av}} = W/\Delta t = 2.70 \times 10^5\,\mathrm{J}/1500\,\mathrm{s}$ $\boxed{P_{\mathrm{av}} = 180\,\mathrm{W}}$
(b) $v_{\mathrm{av}} = 552/1500 = 0.368\,\mathrm{m/s}$, $F_{\parallel} = mg = 490\,\mathrm{N}$:
$P_{\mathrm{av}} = (490)(0.368) = 180\,\mathrm{W}$
Units: $0.180\,\mathrm{kW}$, and $180\,\mathrm{W}/746\,\mathrm{W/hp} = 0.241\,\mathrm{hp}$

## EVALUATE

Order $10^2\,\mathrm{W}$, as predicted, and (a) and (b) agree. Her total output is several times larger: this is only the part that lifts her. ✓

# Example 11: a car at constant speed

PRACTICE




**Key idea**

Constant velocity means $\sum \vec{F} = 0$: the drive force equals the resistive force.

**Problem.** A car moves at a constant $25.0\,\mathrm{m/s}$ on level ground against a total resistive force of $600\,\mathrm{N}$. What power must the engine deliver to the wheels? **Predict:** what is the net force? How do the drive and resistive forces compare?

**IDENTIFY**

Given: $v = 25.0\,\mathrm{m/s}$ (constant), $R = 600\,\mathrm{N}$. Find $P$.

**SET UP**

Constant velocity: $\sum \vec{F} = 0$, so $F = R$ along $\vec{v}$. Use (11) with $\phi = 0^\circ$: $P = Fv$.

**EXECUTE**

Force: $F = R = 600\,\mathrm{N}$

Substitute: $P = Fv = (600\,\mathrm{N})(25.0\,\mathrm{m/s})$

$$\boxed{P = 1.50 \times 10^4\,\mathrm{W} = 15.0\,\mathrm{kW}}$$

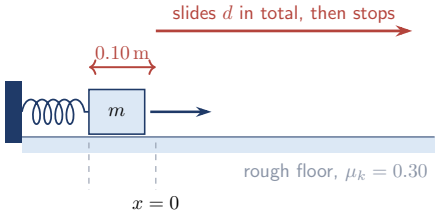
In horsepower: $15\,000\,\mathrm{W}/746\,\mathrm{W/hp} \approx 20\,\mathrm{hp}$

**EVALUATE**

Order $10^4\,\mathrm{W}$, as expected for a car. At the same $R$, doubling $v$ doubles $P$. ✓

*eg. the Accra–Kumasi road.*

# Integrated problem: a block and a spring on a rough floor




## Key idea

No method is named. Decide which forces do work and which states to use.

## Application problem

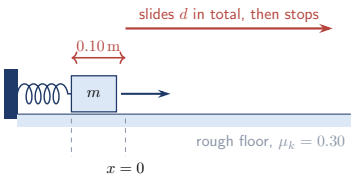
A $0.50\,\mathrm{kg}$ block is pushed against a horizontal spring ($k = 800\,\mathrm{N/m}$), compressing it $0.10\,\mathrm{m}$, and released from rest. The block is not attached to the spring. The floor is rough: $\mu_k = 0.30$ throughout.

(a) How fast is the block moving as it leaves the spring (at natural length)?
(b) How far does it slide in total before it stops?

## Predict first

Is the total distance more or less than $1\,\mathrm{m}$? Sketch, then decide on the states.

# Integrated problem: solution (1)




**Key idea**

Spring force varies, friction is constant: one work–energy equation covers both.

## IDENTIFY

Given: $m = 0.50\,\text{kg}$, $k = 800\,\text{N/m}$, $x_1 = 0.10\,\text{m}$, $v_1 = 0$, $\mu_k = 0.30$. Find: (a) $v$ at $x = 0$; (b) total distance $D$.

## SET UP

Forces: spring (varies, (9b)), friction (constant, $-fd$), $\vec{\boldsymbol{n}}$ and $\vec{\boldsymbol{w}}$ (perpendicular). States: 1 released; 2 at $x = 0$; 3 at rest. Use (5).

*Reuse:* (1) friction work, (9b) spring work, (5) theorem

## EXECUTE: the common pieces

Friction: $f = \mu_k mg = (0.30)(0.50\,\text{kg})(9.8\,\text{m/s}^2) = 1.47\,\text{N}$

Spring: $W_{\text{by}} = \frac{1}{2}kx_1^2 = \frac{1}{2}(800\,\text{N/m})(0.10\,\text{m})^2 = 4.00\,\text{J}$

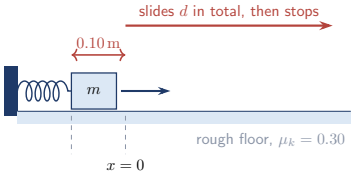
## EXECUTE: (a) states 1 to 2

Friction over $0.10\,\text{m}$: $W_f = -(1.47\,\text{N})(0.10\,\text{m}) = -0.147\,\text{J}$

Total: $W_{\text{tot}} = 4.00\,\text{J} - 0.147\,\text{J} = 3.85\,\text{J} = \frac{1}{2}mv_2^2$, so

$$v_2 = \sqrt{\frac{2(3.85\,\text{J})}{0.50\,\text{kg}}}$$

$$\boxed{v_2 = 3.9\,\text{m/s}}$$




**Key idea**

Friction removes all $4.00\,\text{J}$ the spring gave: $fD = \frac{1}{2}kx_1^2$.

## EXECUTE: (b) states 1 to 3

Friction acts over the whole path $D$ from release to rest. The spring does $4.00\,\text{J}$.

Equation: $W_{\text{tot}} = \frac{1}{2}kx_1^2 - fD = 0 - 0$, so $D = \dfrac{\frac{1}{2}kx_1^2}{f}$

Substitute: $D = \dfrac{4.00\,\text{J}}{1.47\,\text{N}}$ $\boxed{D = 2.7\,\text{m}}$ (total, from release)

## EVALUATE

More than $1\,\text{m}$, as predicted. Check: after leaving the spring the block slides $2.72 - 0.10 = 2.62\,\text{m}$, and friction does $-(1.47)(2.62) = -3.85\,\text{J}$, exactly removing the kinetic energy found in (a). ✓

## MODIFY

What if $\mu_k$ were doubled? Then $f = 2.94\,\text{N}$ and $D$ halves to $1.4\,\text{m}$. What if the block were attached to the spring? It would pull back and oscillate about $x = 0$, and the answer to (b) would change.

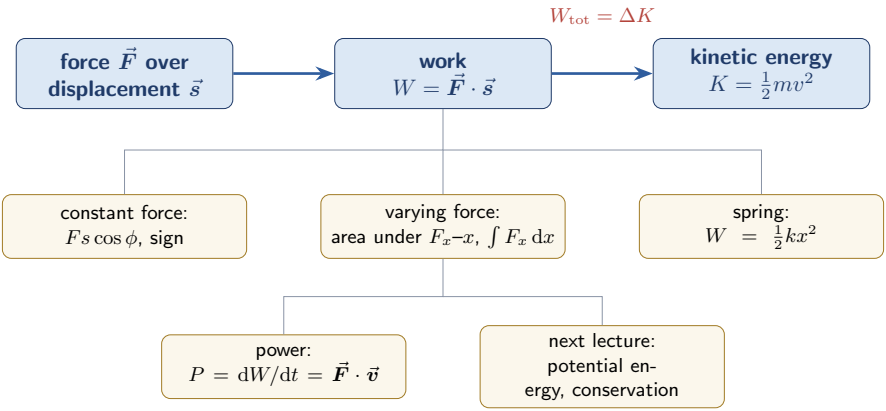
## Common mistakes and the correct habit

| Mistake | Instead |
|---|---|
| Using $W = Fs\cos\phi$ with a force that varies. | Integrate $F_x$ over $x$, or take the area. |
| Measuring the spring's stretch from the wrong point. | From the natural length. |
| Mixing the signs of the work by and on the spring. | $W_{\text{by}} = -W_{\text{on}}$. |
| Putting time limits in a work integral. | The limits are positions $x_1$ and $x_2$. |
| Confusing power with energy. | Power is a rate in watts; a kilowatt-hour is energy. |
| Counting work done by a perpendicular force. | It is zero at every point. |

**But what if...** the energy lost to friction does not vanish? It becomes thermal energy. That bookkeeping is Lecture 5 and the thermodynamics lectures.

# Week 4 concept map

CORE




Work connects a force over a displacement to the change in kinetic energy. The branches are the three kinds of force you met: constant, varying and spring.

## Key ideas

- **Only the part of the force along the displacement does work: $W = \vec{F} \cdot \vec{s} = Fs\cos\phi$. A perpendicular force, or no displacement, means no work.**
- The sign of $W$ is the sign of $\cos\phi$: positive speeds an object up, negative slows it down. Work is a scalar.
- Total work is the signed sum of the works, or $(\sum \vec{F}) \cdot \vec{s}$. Use one method and check with the other.
- **The net work changes the kinetic energy: $W_{\text{tot}} = \Delta K$, with $K = \frac{1}{2}mv^2$. $K$ is a scalar and never negative.**
- Doubling the speed multiplies $K$ by $4$. This is why stopping distance grows as $v^2$.
- The energy method gives speeds, distances and forces without time. Use Newton's second law for accelerations and times.
- **For a varying force, work is the area under $F_x$–$x$, or $\int F_x\,\mathrm{d}x$. For a spring, $W_{\text{on}} = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2$.**
- Power is the rate of doing work: $P = W/\Delta t$ and $P = \vec{F} \cdot \vec{v}$. A kilowatt-hour is energy.
- **Habits:** sketch, predict the sign of $W_{\text{tot}}$, calculate, check units and the prediction.

**Reference tools:** the next slides (equation map, master toolkit). The Reference deck holds the seven toolkits and the equation-anatomy slides.

# Reference tools: the equation map

| Idea | Equation | Conditions and units |
|---|---|---|
| **(1) Work** | $W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}} = Fs\cos\phi$ | constant $\vec{\boldsymbol{F}}$, straight $\vec{\boldsymbol{s}}$; J |
| **(2) Components** | $W = F_x s_x + F_y s_y$ | same conditions |
| **(3) Total work** | $W_{\text{tot}} = \sum W_i = (\sum \vec{\boldsymbol{F}}) \cdot \vec{\boldsymbol{s}}$ | signed sum |
| **(4) Kinetic energy** | $K = \frac{1}{2}mv^2$ | scalar, $\geq 0$; J |
| **(5) Work–energy** | $W_{\text{tot}} = K_2 - K_1$ | all forces; particle |
| **(6) Gravity** | $W_w = \pm mgh$ | down $+$, up $-$; $g$ constant |
| **(7) Varying force** | $W = \int_{x_1}^{x_2} F_x \, \mathrm{d}x$ | area under $F_x$–$x$; limits are positions |
| **(8) Hooke's law** | $F_x = kx$ | small stretch; N/m |
| **(9) Spring work** | $W_{\text{on}} = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2, \quad W_{\text{by}} = -W_{\text{on}}$ | $x$ from natural length; J |
| **(10), (11) Power** | $P_{\text{av}} = W/\Delta t, \quad P = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{v}}$ | W |

## • MASTER TOOLKIT – Work and energy

**1. IDENTIFY** The object, and what is asked: a speed, a distance, a force, a work or a power.
**2. STATES** Mark the initial and final positions with their speeds.
**3. FBD** All forces on the object, and $\vec{s}$ (Week 3).
**4. WORK OF EACH FORCE** Sign first, then size.
**5. PREDICT** Sign of $W_{\text{tot}}$, speeding up or slowing down, rough size.
**6. METHOD** Energy, or Newton's second law.
**7. EXECUTE** $W_{\text{tot}} = \Delta K$; add power if a rate is asked.
**8. EVALUATE** Units, sign of $\Delta K$, check by a second method.

## Step 4: the key move for each force

**Constant force:** $W = Fs\cos\phi$; find $\phi$ between $\vec{F}$ and $\vec{s}$.
**Perpendicular force:** $W = 0$.
**Friction:** $W_f = -f_k d$ while sliding.
**Gravity:** $W_w = \pm mgh$, from the height change only.
**Varying force:** area under $F_x$–$x$ or $\int F_x \, \mathrm{d}x$.
**Spring:** $W_{\text{by}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$.
**Power:** $P = W/\Delta t$ or $P = Fv\cos\phi$.

**Framework:** IDENTIFY (1–2), SET UP (3–6), EXECUTE (7), EVALUATE (8).

# Where this goes next

## Lecture 5: potential energy

The work done by gravity and by springs does not depend on the path, so it can be stored as potential energy. $W_{\text{tot}} = \Delta K$ then becomes conservation of energy.

## Lecture 8: rotation

Rotational work $W = \tau\,\Delta\theta$ and power $P = \tau\omega$ mirror $W = Fs$ and $P = Fv$.

## Lectures 9 to 11: heat and thermodynamics

The kinetic energy removed by friction appears as thermal energy. The work done by a gas, $\int p\,\mathrm{d}V$, is the same integral as $\int F\,\mathrm{d}x$.

## Earlier weeks, now working together

Week 1's dot product gives the work. Week 2's area under a graph gives varying work, and $v^2 = v_0^2 + 2a\Delta x$ gives the theorem. Week 3's free-body diagram says which forces to list.

## Physical meaning

Energy is the central bookkeeping idea of the course. The same $W_{\text{tot}} = \Delta K$ method returns in every remaining lecture.

# Checklist: can I . . . CORE

- ☐ calculate work as $\vec{F} \cdot \vec{s} = Fs\cos\phi$?
- ☐ decide the sign of work from the angle?
- ☐ list the forces on a free-body diagram that do work?
- ☐ find total work by two methods?
- ☐ state the work–energy theorem and use $K = \frac{1}{2}mv^2$?
- ☐ explain why $K \propto v^2$?
- ☐ choose between the energy method and Newton's second law?
- ☐ find work from an $F_x$–$x$ graph or an integral?
- ☐ calculate the work done by or on a spring?
- ☐ explain why a perpendicular force does no work?
- ☐ calculate average and instantaneous power?
- ☐ select the appropriate model in an unfamiliar problem?

## Next

Tutorial sheet: extra work, work–energy, spring, friction and power problems, the push-down box, the hard workout and the stool that slides to rest (section C, solved), and a ladder from independent practice to exam-style items. The Reference deck holds the supporting material cut from the live lecture. The supplementary notes hold the dot-product and calculus refreshers, the proof of the theorem for variable forces, work along a general path, and what happens to the energy lost to friction.

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